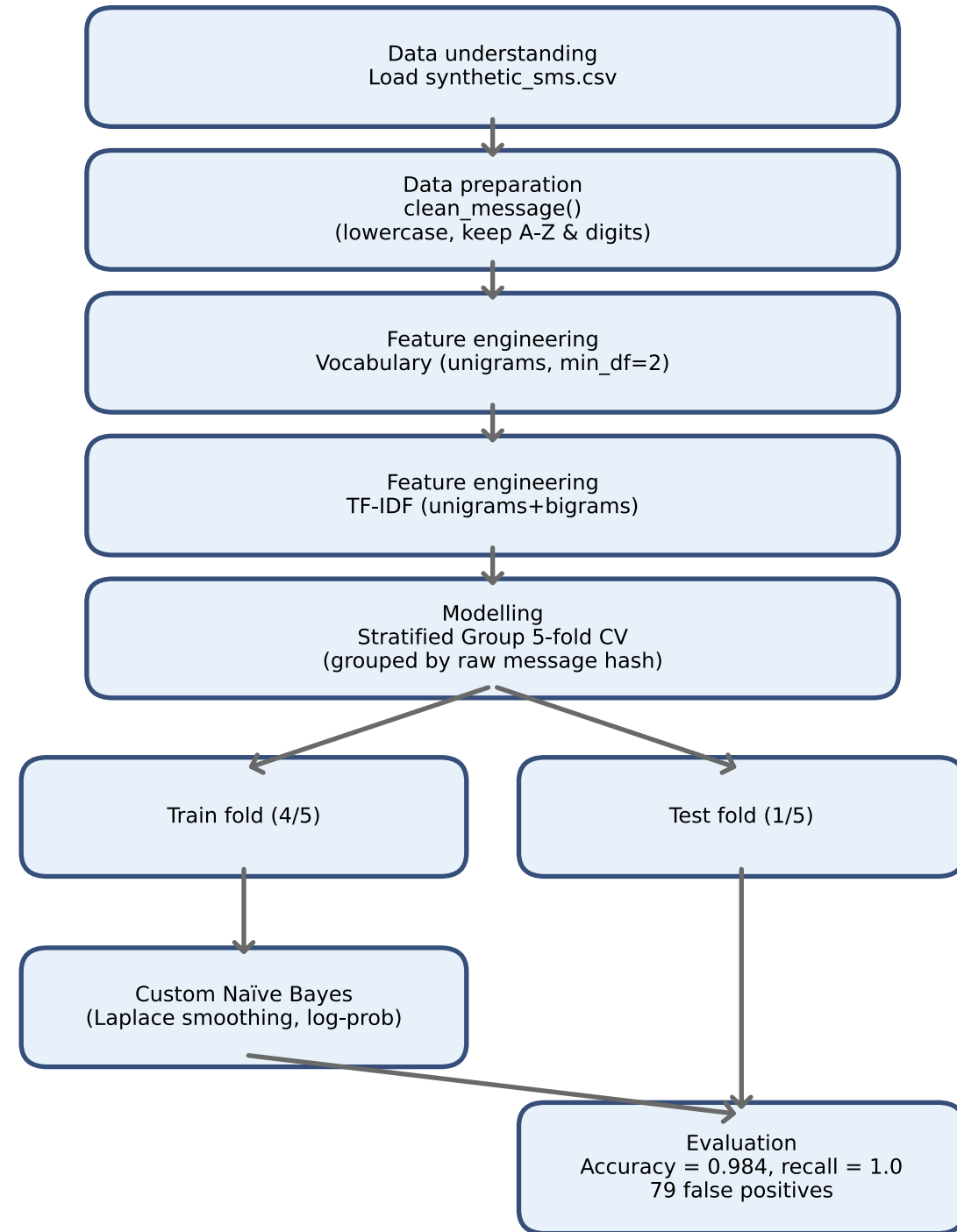


CENTRALISED PIPELINE (Custom NB, TF-IDF, Stratified Group K-Fold)



FEDERATED LEARNING SIMULATION (Bag-of-words counts, local differential privacy)

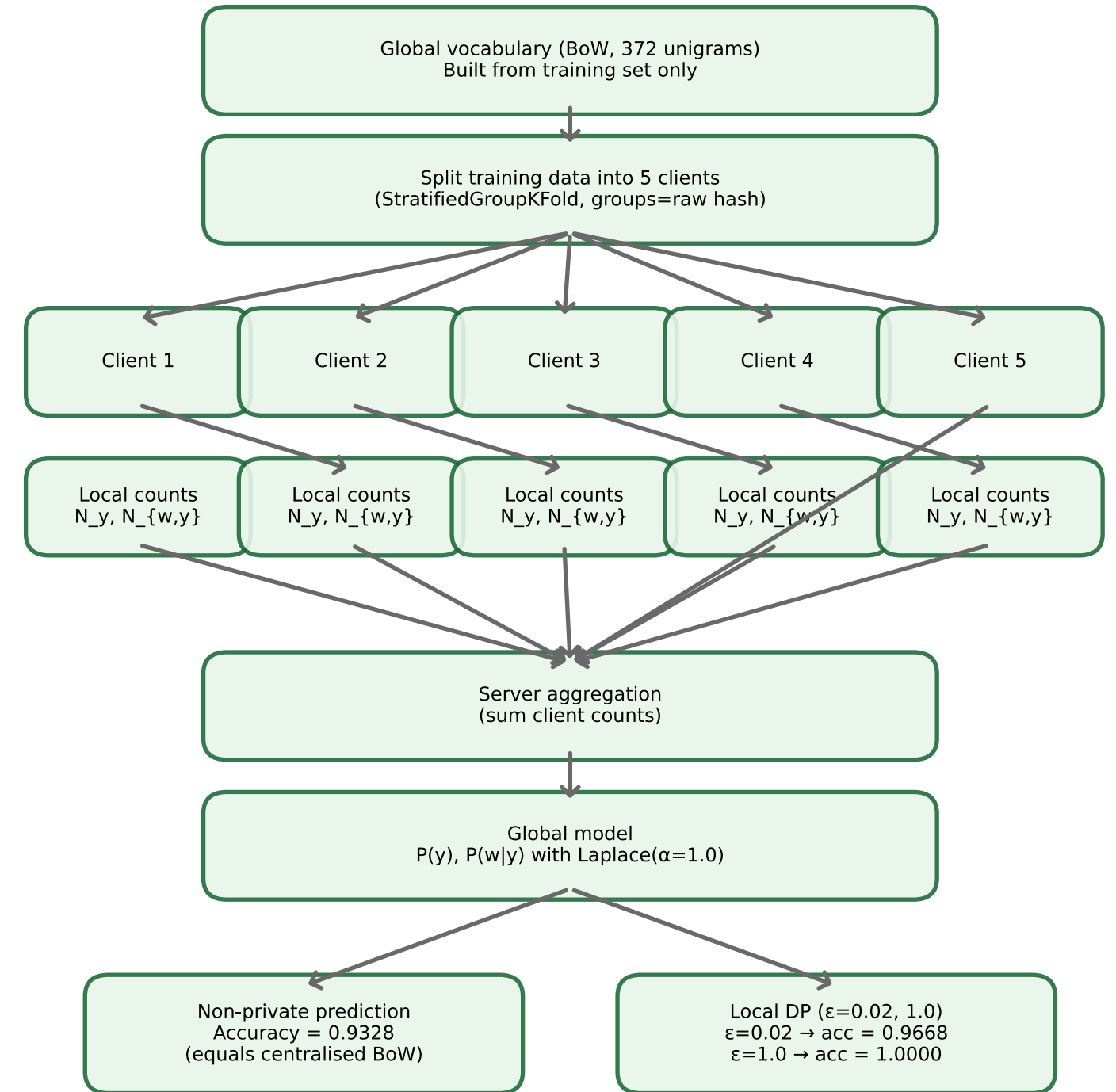


Figure 1: Implementation pipelines – Centralised (left) and Federated (right). Centralised uses TF-IDF with grouped cross-validation and a custom NB classifier. Federated uses bag-of-words counts with per-client local aggregation and optional DP.